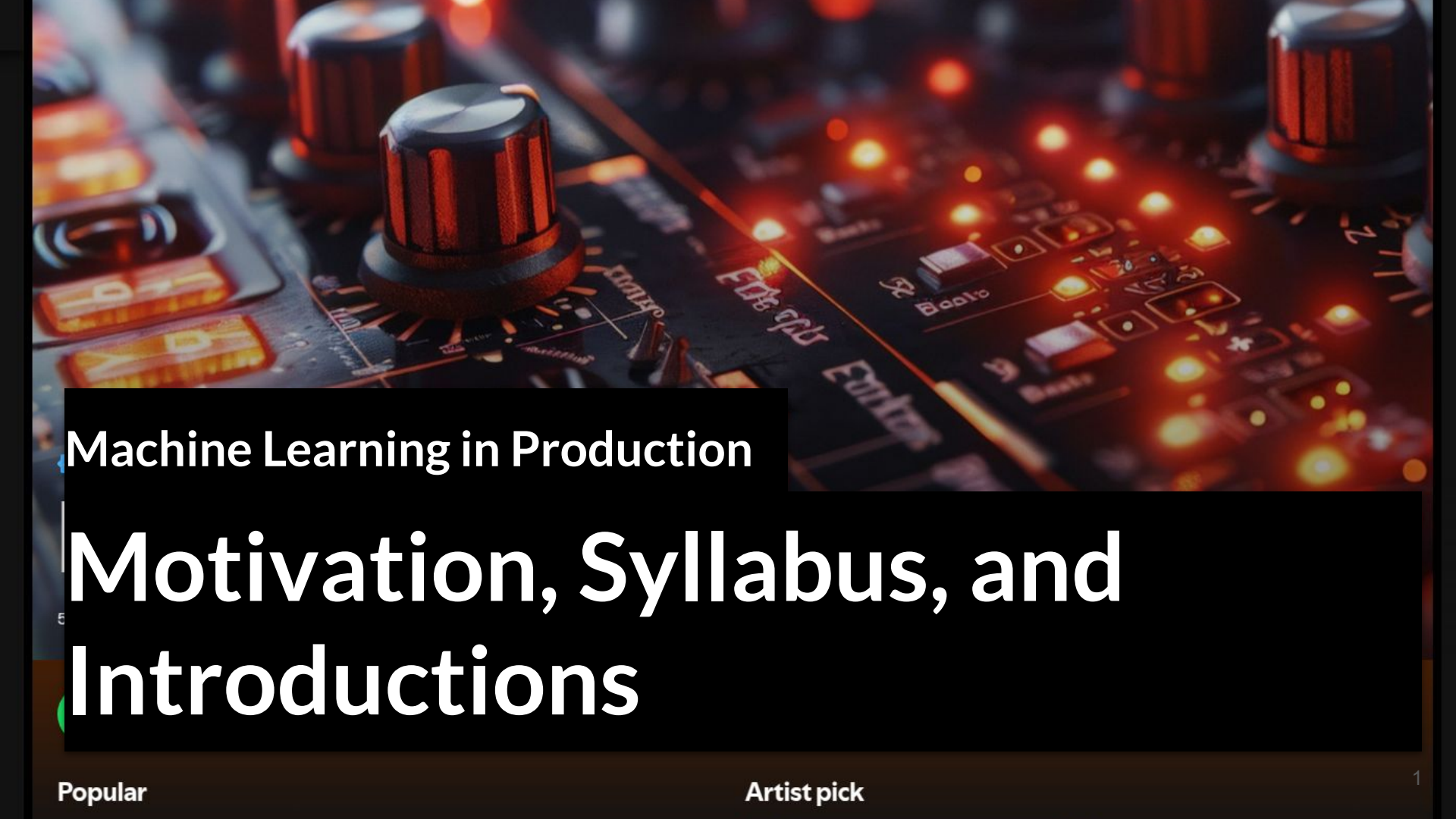




Machine Learning in Production

Motivation, Syllabus, and Introductions



Verified Artist

Machine Learning in Production

Motivation and Introduction

Popular

Artist pick

Slack

- We use Slack for this course, including during lectures
- See signup link on Canvas (you can do this right now!)
- Setup the ability to read/post to Slack during lecture.

Introductions

Before the next lecture, introduce yourself in Slack channel #social:

- Your (preferred) name
- In 1-2 sentences, describe your data science background and goals (e.g., coursework, internships, work experience)
- In 1-2 sentences, describe your software engineering background, if any, and goals (e.g., coursework, internships, work experience)
- The kind of job you are aiming for after graduation
- A hobby or a favorite activity outside school

Waitlist situation

We let people in as spaces become available, but will likely not clear all waitlist. Expect more movement on 17645 than 11695

Please email the course admin Jenni Cooper (cooperj@andrew.cmu.edu) 17*45 or Karen Sue Kirk (karensuk@andrew.cmu.edu) for 11695, not us, to inquire about WL status/registration movement in.

If and when you join late:

1. Post an introduction on Slack (#social)
2. See the Canvas front page about automatic extensions

Learning Goals

- Understand how ML components are parts of larger systems
- Illustrate the challenges in engineering an ML-enabled system beyond accuracy
- Explain the role of specifications and their lack in machine learning and the relationship to deductive and inductive reasoning
- Summarize the respective goals and challenges of software engineers vs data scientists
- Explain the concept and relevance of “T-shaped people”
- Understand the basic mechanics of this course.

Case Study: Music Generation

Context: Music Generation Research

Lam, Max WY, et al. "Efficient neural music generation." Advances in Neural Information Processing Systems 36 (2024).

Recent progress in music generation has been remarkably advanced by the state-of-the-art MusicLM, which comprises a hierarchy of three LMs, respectively, for semantic, coarse acoustic, and fine acoustic modelings. Yet, sampling with the MusicLM requires processing through these LMs one by one to obtain the fine-grained acoustic tokens, making it computationally expensive and prohibitive for a real-time generation. Efficient music generation with a quality on par with MusicLM remains a significant challenge. In this paper, we present MeLoDy (M for music; L for LM; D for diffusion), an LM-guided diffusion model that generates music audios of state-of-the-art quality meanwhile reducing 95.7% to 99.6% forward passes in MusicLM, respectively, for sampling 10s to 30s music. MeLoDy inherits the highest-level LM from MusicLM for semantic modeling, and applies a novel dual-path diffusion (DPD) model and an audio VAE-GAN to efficiently decode the conditioning semantic tokens into waveform. DPD is proposed to simultaneously model the coarse and fine acoustics by incorporating the semantic information into segments of latents effectively via cross-attention at each denoising step. Our experimental results suggest the superiority of MeLoDy, not only in its practical advantages on sampling speed and infinitely continuable generation, but also in its state-of-the-art musicality, audio quality, and text correlation. Our samples are available at <https://Efficient-MeLoDy.github.io/>.

The Startup Idea

You just completed a research thesis as part of your Master's/PhD degree about making deep learning for generative AI more energy efficient.

Your recent project was using music generation as a case study. You showed 30% energy improvements with similar quality of generated music on benchmark prompts.

Two friends are excited about the application of music generation.

Idea: Let's commercialize the idea and sell it to end users or music producers.

Suno Showcase

- Home
- Create
- Library
- Explore
- Search

2:33

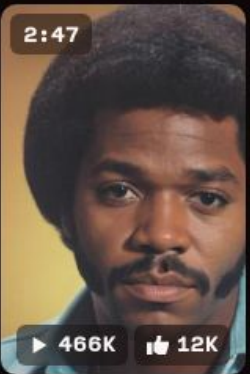


▶ 263K 👍 3.2K

Suno
yolkhead, 🌱, 3daisy, Bru...

 yolkhead ⋮

2:47



▶ 466K 👍 12K

Ain't Got a Nickel Ain't...
up tempo Memphis soul ...

 Soul_Diego ⋮

2:48



▶ 181K 👍 2.2K

Vapor of Feelings -...
Italo disco, Italo-disco, 19...

 midicheat ⋮

3:15



▶ 681K 👍 6.8K

Give it to me (Suno) 🌱

Electronic, Drum and Ba...

 Mr.Tree 🌱 ⋮

3:23



▶ 126K 👍 1.1K

Butterflies
Winding, Introspective, m...

 GTZY ⋮

4:24



▶ 434K

I Can Wait
haunted ma...

 wetco ⋮

Trending Songs

3:08



2:58



4:26



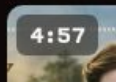
3:06



3:05



4:57





beatsbyaiofficial Beats By Ai

Follow

Message



29 Following 326.4K Followers 10.6M Likes

For Full Songs Click The Link! 📌

🔗 linktr.ee/beatsby.ai

📺 Videos

↻ Reposts

👍 Liked



Verified Artist

Beats By AI

592,469 monthly listeners



Follow



Popular

- | | | | | |
|---|--|--------------------------|-----------|------|
| 1 | | Country Girls Make Do | 1,413,135 | 2:29 |
| 2 | | Only Thing I Love | 936,465 | 1:54 |
| 3 | | Sloppy Top At The Movies | 861,948 | 2:55 |

Artist pick



Breakout: Likely challenges in building commercial product?

As a group, think about challenges that the team will likely face when turning their research into a product:

- One machine-learning challenge
- One engineering challenge in building the product
- One challenge from operating and updating the product
- One team or management challenge
- One business challenge
- One safety or ethics challenge

Post answer to #lecture on Slack and tag all group members (skip if nobody in group has slack set up yet)

Examples for discussion

- What does correctness or accuracy really mean? What accuracy do customers care about?
- How can we see how well we are doing in practice? How much feedback are customers going to give us before they leave?
- Can we estimate how good our generated music is? How are we doing for different customers or different genres?
- How to present results to the customers?
- When customers complain about poor output, how to prioritize and what to do?
- What are unacceptable mistakes and how can they be avoided? Is there a safety risk?
- Can we cope with an influx of customers?
- Will responding to the same prompt twice produce the same result? Does it matter?
- How can we debug and fix problems? How quickly?

Examples for discussion 2

- With more customers, generation is taking longer and longer – what can we do?
- Generation sometimes crashes. What to do?
- How do we achieve high availability?
- How can we see that everything is going fine and page somebody if it is not?
- We improve our model but somehow system behavior degrades... Why?
- Tensorflow update; does our infrastructure still work?
- Once somewhat successful, how to handle large amounts of data per day?
- Buy more machines or move to the cloud?
- Models are continuously improved. When to deploy? Can we roll back?
- Can we offer live music generation as an app? As a web service?
- Can we get better the longer a user interacts with us? Will this benefit the next song they generate as well?

Examples for discussion 3

- How many genres can be supported? Do we have the server capacity?
- How specific should genres be?
- How to make it easy to support new types of music?
- Can we generate different types of vocals? Handle different languages or accents?
- Can and should we learn from customer data?
- How can we debug problems in files we are not allowed to see?
- Any chance we might leak private data?
- Can competitors or bad actors attack our system?

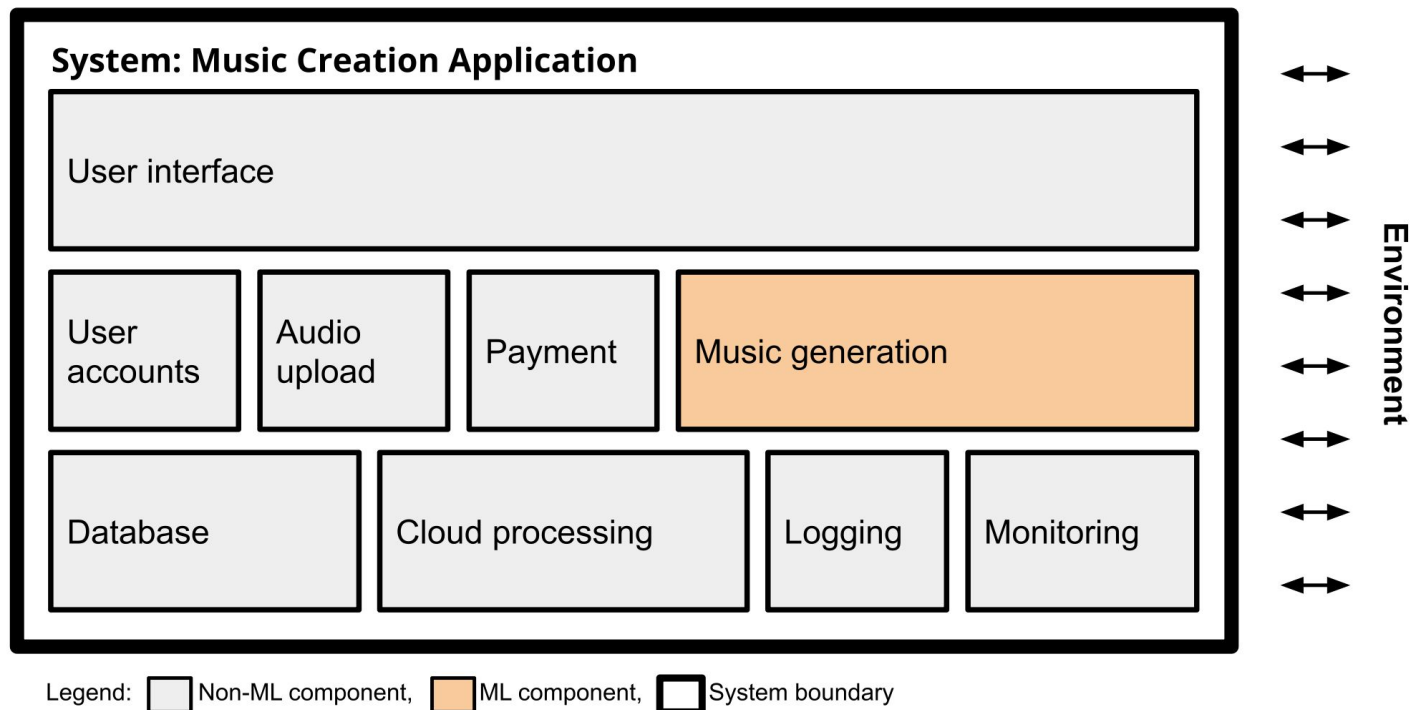
What qualities are important for a good commercial music generator?



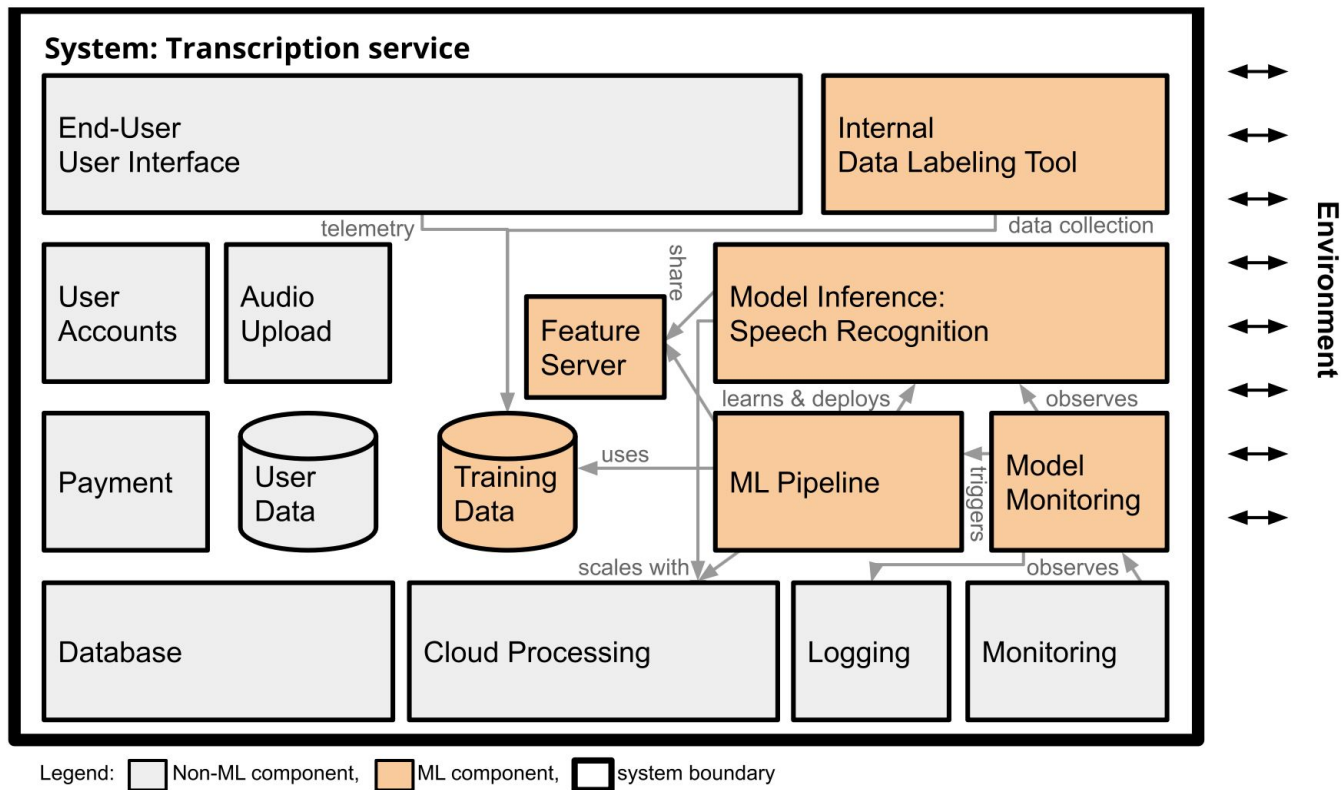
What non-ML components are needed?



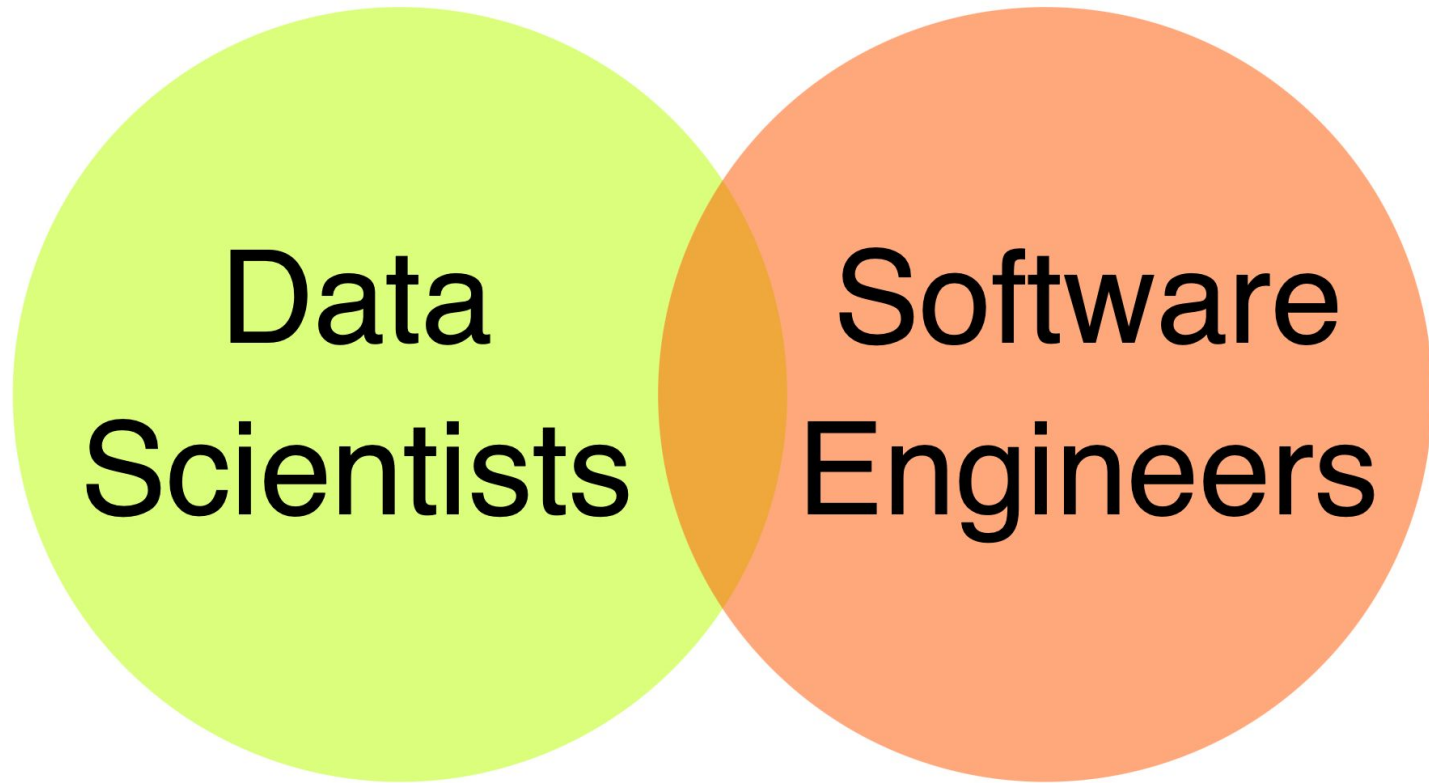
ML in a Production System



ML in a Production System



Development Teams



and Data engineers + Domain specialists + Operators + Business team + Project managers + Designers, UI Experts + Safety, security specialists + Lawyers + Social Scientists + ...

Data scientist

- Often fixed dataset for training and evaluation (e.g., PBS interviews)
- Focused on accuracy
- Prototyping, often Jupyter notebooks or similar
- Expert in modeling techniques and feature engineering
- Model size, updateability, implementation stability typically does not matter

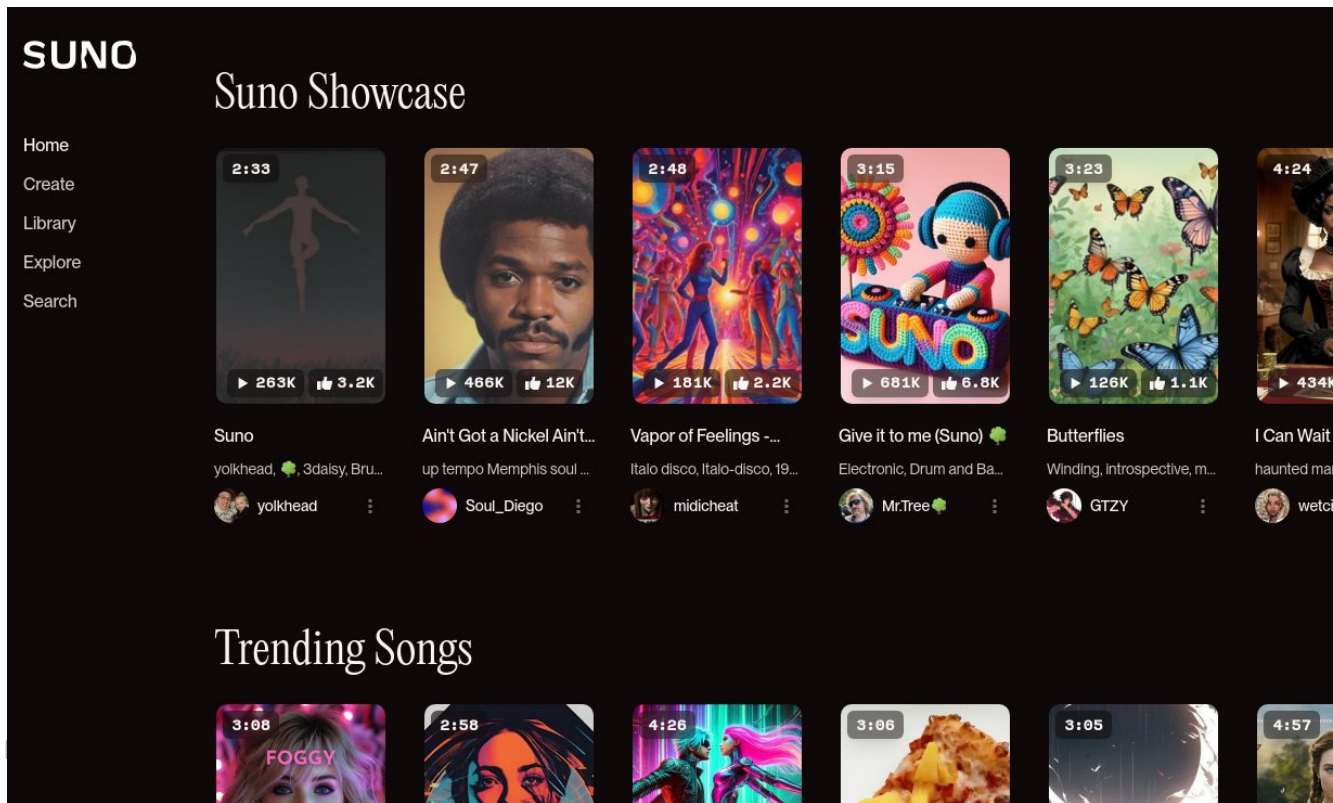
Software engineer

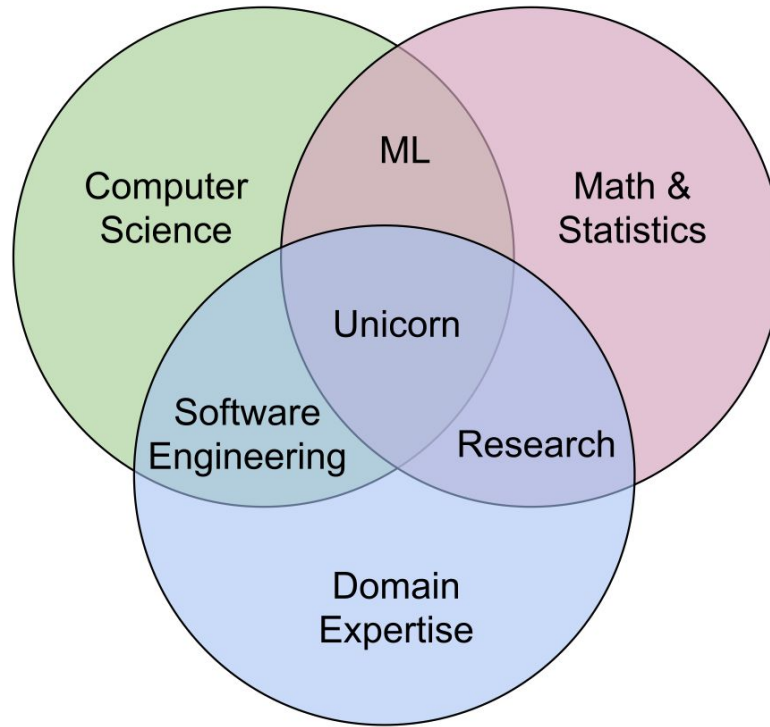
- Builds a product
- Concerned about cost, performance, stability, release time
- Identify quality through customer satisfaction
- Must scale solution, handle large amounts of data
- Detect and handle mistakes, preferably automatically
- Maintain, evolve, and extend the product over long periods
- Consider requirements for security, safety, fairness

Likely collaboration challenges?



What might Software Engineers and Data Scientists focus on?





By Steven Geringer, via Ryan Orban. Bridging the Gap Between Data Science & Engineer: Building High-Performance Teams. 2016

T-Shaped People

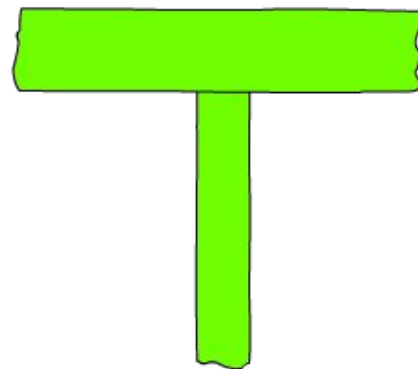
Broad-range generalist + Deep expertise



"I-shaped"
Expert at one thing



Generalist
Capable in a lot of things
but not expert in any



"T-shaped"
Capable in a lot of things
and expert in one of them

Figure: Jason Yip. Why T-shaped people?. 2018

T-Shaped People

Broad-range generalist + Deep expertise

Example:

- Basic skills of software engineering, business, distributed computing, and communication
- Deep skills in deep neural networks (technique) and medical systems (domain)

Latest Buzzword: π -Shaped People



Syllabus and Logistics

Syllabus and Class Structure

17-445/17-645/17-745, Spring 2026, 12 units

Monday/Wednesdays 2:00-3:20pm

Recitation Fridays 9:30am, 11am, 2pm

Communication

Preferably ping us on Slack (link on Canvas)

Email also okay, but ask your chatbot to keep the email short

All announcements through Slack #announcements

Weekly office hours, starting next week, schedule on Canvas

Post questions on Slack – Please use #general or #assignments and post publicly if possible; your classmates will benefit from your Q&A!

All course materials (slides, recordings, assignments, old midterms) available on course website: <https://mlip-cmu.github.io/s2026/>



Office Hours

Times posted soon on Canvas

Feel free to drop in with questions or just to chat

Later office hours will have signup sheet for homework check-ins

Class with software engineering flavor

Focusing on engineering judgment

Arguments, tradeoffs, and justification, rather than single correct answer

Practical engagement, building systems, testing, automation

Strong teamwork component

Both text-based and code-based homework assignments



Fundamentals of Engineering AI-Enabled Systems

Holistic system view: AI and non-AI components, pipelines, stakeholders, environment interactions, feedback loops

Requirements:

System and model goals
User requirements
Environment assumptions
Quality beyond accuracy
Measurement
Risk analysis
Planning for mistakes

Architecture + design:

Modeling tradeoffs
Deployment architecture
Data science pipelines
Telemetry, monitoring
Anticipating evolution
Big data processing
Human-AI design

Quality assurance:

Model testing
Data quality
QA automation
Testing in production
Infrastructure quality
Debugging

Operations:

Continuous deployment
Contin. experimentation
Configuration mgmt.
Monitoring
Versioning
Big data
DevOps, MLOps

Teams and process: Data science vs software eng. workflows, interdisciplinary teams, collaboration points, technical debt

Responsible AI Engineering

Provenance,
versioning,
reproducibility

Safety

Security and
privacy

Fairness

Interpretability
and explainability

Transparency
and trust

Ethics, governance, regulation, compliance, organizational culture

Key Changes this Semester

Public lecture recordings

Updates to several homework assignments and to the group project

Required, graded in-person interaction with course staff after every assignment

Opt-in participation grading

10% penalty on resubmissions for tokens

No more reading quizzes

Thoughts on Generative AI for Homework?



ChatGPT, Copilot, Claude Code, ...? Homework, labs, midterms, ...?

Our Position on Generative AI for Homework

This is a course on responsible building of ML products. This includes questions of how to build generative AI tools responsibly and discussing what use is ethical.

Feel free to use them and explore whether they are useful.

We assume you are using AI coding tools for assignments

Warning: Be aware of hallucinations and AI slop. Requires understanding to check answers. We test them; they often need extra hand-holding or generate bad/wrong answers. (Last semester 80% of students lost 20+% points on individual assignment due to not critically checking AI output of I3)

You are responsible for the correctness of what you submit!

Our Position on Generative AI for Homework II

Even if you generate correct solutions, we expect you to understand them

We gave up on written answers for “reflection” questions

Every assignment has a requirement to explain your solution to the course staff and to answer questions probing your understanding

(You can attempt this multiple times if needed)

No AI use in midterm

Prerequisites

Some machine learning experience required

- Basic understanding of data science process, incl. data cleaning, feature engineering, using ML libraries
- High level understanding of machine-learning approaches
 - Supervised learning
 - Regression, decision trees, neural networks
 - Accuracy, recall, precision, ROC curve
- Ideally, some experience with notebooks, sklearn, or other frameworks

Basic programming and command-line skills will be needed

No further software-engineering knowledge required

- Teamwork experience in product team is useful but not required
- No required exposure to requirements, software testing, software design, continuous integration, containers, process management, etc.
 - If you are familiar with these, there will be some redundancy – sorry!

First Homework Assignment I1

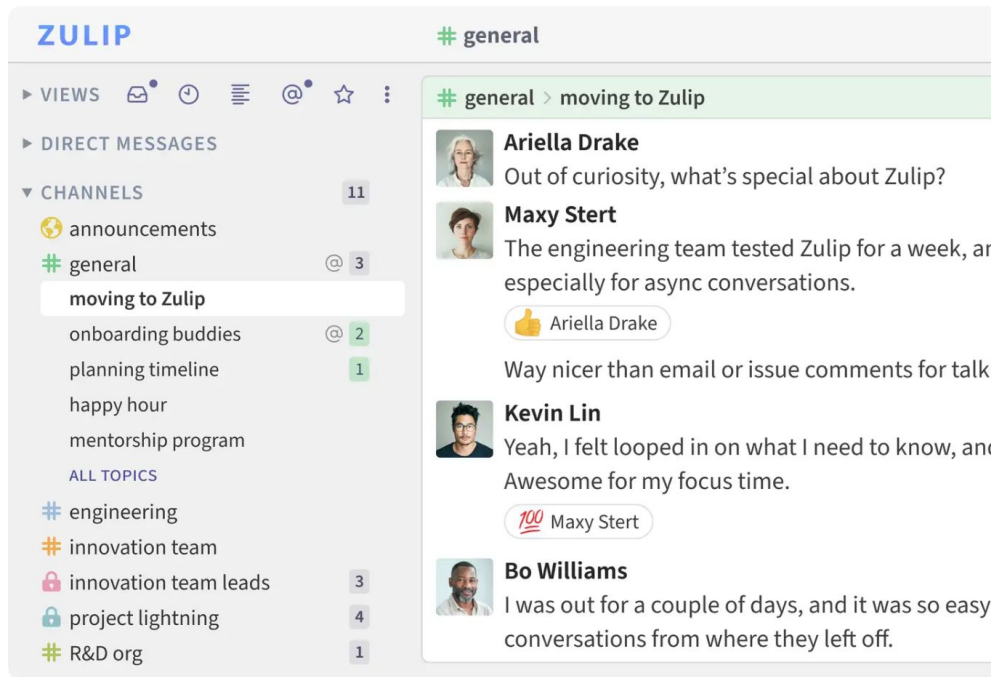
“Coding warmup assignment”

Out tonight, due Monday Jan 26

Enhances Slack-style mature web application with two LLM-powered features

Coding assignment in nontrivial code base, front-end and backend, change existing code, learn new APIs, handle credentials

Use of AI coding tools encouraged



Active lecture

Case study driven

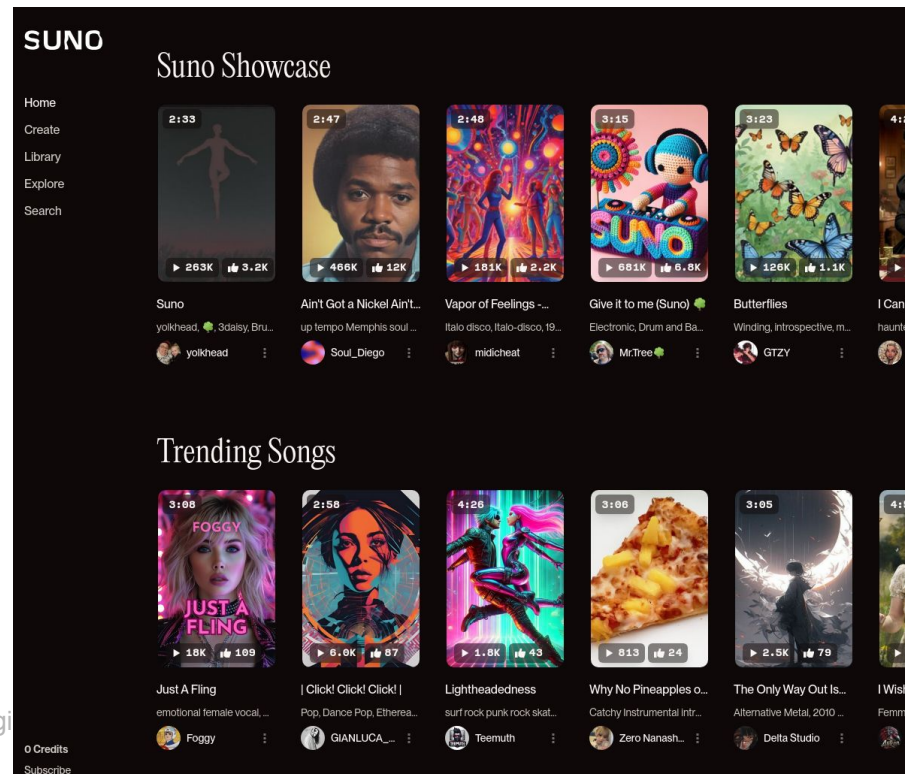
Discussions highly encouraged

Regular in-class activities, breakouts

Contribute your own experience!

Discussions over definitions

Machine Learning in Production • Claire Le Goues & Christian Kaestner Carnegie Mellon University



Recordings and Attendance

Strongly recommend to attend lecture – discussions are important to learning

[insert warning about relying on recordings]

Participation can be part of your grade (opt-in before first midterm)

Best effort lecture recordings, textbook, and slides available

Contact us for accommodations (illness, interview travel, unforeseen events), or have your advisor reach out. We try to be flexible.

Readings

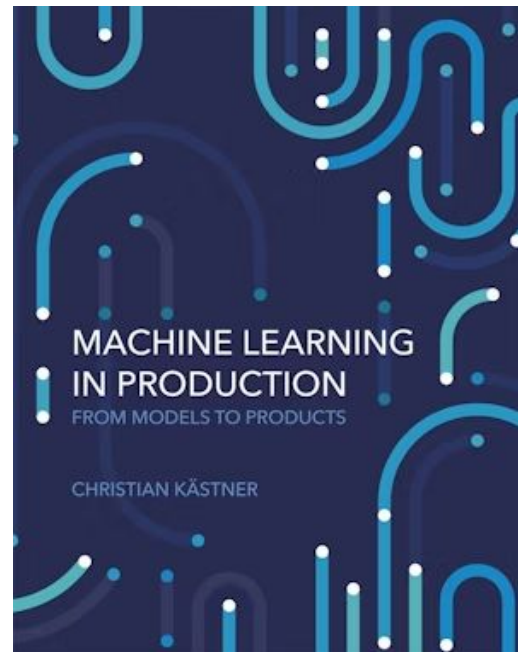
We will selectively assign readings (papers, blog posts, podcasts), about 1 per week

You will discuss/apply readings during the lecture

We might ask about readings content in the midterm

Optional: Supplementary textbook, closely tracks lecture content.

Published online at <https://mlip-cmu.github.io/book/>,
printed through MIT Press



Grading

- 35% individual assignment
- 30% group project with final presentation
- 25% two in-class exams (or 15% if opting in to graded participation)
- 0% participation (or 10% if opting in to graded participation)
- 10% labs
- No final exam (final presentations will take place in that time slot)

Expected grade cutoffs in syllabus (>82% B, >94 A-, >96% A, >99% A+)

Assignments

Series of 4 small to medium sized **individual assignments**

- Build LLM-powered features for chat application
- Create a LLM-powered hazard analysis tool and apply it for a smart dashcam system
- Secure an MCP AI agent for customer service support
- Create end-user explanations for a medical device

Large **team project** with 4 milestones:

- Build and deploy a prediction (movie recommendation) service
- Testing in production, monitoring
- Final presentation

Usually due Monday night; see schedule

Research in this Course

We are conducting academic research in this course (I1 and I4).

This research will involve analyzing student work of assignment **after the end of the semester.**

You will not be asked to do anything above and beyond the normal learning activities and assignments that are part of this course. All data will be analyzed in de-identified form and presented in the aggregate, without any personal identifiers.

You are free not to participate in this research, and your participation will have no influence on your grade for this course or your academic career at CMU. If you do not wish to participate, you can opt out (see instructions in assignments); instructors will not know who opts out before assigning final grades.

17-745 PhD Research Project

Research project instead of individual assignments I3 and I4

Design your own research project and write a report

- A case study, empirical study, etc.

Very open ended: Align with own research interests and existing projects; requires preliminary results and final presentation

See the [project requirements](#) and deadlines and talk to us

Midterms and Final Presentation

2 Midterms, pen&paper during lecture time, not cumulative

(find old midterms as examples on the course website)

No written final exam, but graded presentations in final exam slot

Date announced February 16:

<https://www.cmu.edu/hub/docs/final-exams.pdf>

May be as late as May 4; do not book flights early

Labs

*Please attend the lab for which you are registered
(section G received special instructions).*

Introducing various tools, e.g., fastAPI (serving), Kafka (stream processing), Jenkins (continuous integration), MLflow (experiment tracking), Docker & Kubernetes (containers), Prometheus & Grafana (monitoring), SHAP (explainability)...

Hands on exercises, bring a laptop

Often introducing tools useful for assignments

About 1h of work, graded pass/fail, low stakes, show & explain work to TA

First lab **this Friday**: Calling, securing, and creating APIs

Lab grading and collaboration

We recommend that you start each lab before the recitation, but it can be completed during

Graded pass/fail by TA on the spot, can retry

Relaxed collaboration policy: Can work with others before and during recitation, but have to show & explain solution to TA individually

(Think of recitations as mandatory office hours)

Lab online options (experimental)

If you complete the lab ahead of time, you may be able to show it over Zoom on Thursday evenings

You will show/explain your work to the TA over Zoom

TA will grade but not provide help; if you like help, attend your lab session in person on Friday

Logistics on how and when to queue will be posted on Slack

Participation

Participation != Attendance

Expecting active participation at minimum in breakout discussions

When not feeling well—please stay home and get well, and email us for accommodation

Grading (if opted in):

- 100%: Participates actively at least once in most lectures (4 lectures waived, no questions asked)
- 80%: Participates actively at least once in two thirds of the lectures
- 60%: Participates actively at least once in over half of the lectures
- 30%: Participates actively at least once in one quarter of the lectures
- 0%: Participation in less than one quarter of the lectures.

Grading Philosophy

Specification grading, based in adult learning theory

Giving you choices in what to work on or how to prioritize your work

We are making every effort to be clear about expectations (specifications), will clarify if you have questions

Assignments broken down into expectations with point values, each graded **pass/fail**

Opportunities to resubmit work until last day of class

[\[Example\]](#)

Token System for Flexibility

8 individual tokens per student:

- Submit individual assignment 1 day late for 1 token (after running out of tokens, 15% penalty per late day)
- Redo individual assignment for 3 token and a 10% tax
- Late/redo lab, team feedback, ... for 1 token
- Extra participation day for 1 token

8 team tokens per team:

- Submit milestone 1 day late for 1 token
- Redo milestone for 3 token and a 10% tax

How to use tokens

No need to tell us if you plan to submit very late. We will assign 0 and you can resubmit

Resubmit through Canvas

Request token use for late lab during office hours

#remaining tokens reflected on Canvas

Accommodations

The token system provides built-in flexibility and will cover many issues

Slack/email for accommodations, like extensions or waived participation

- Usually exceptional, unanticipated circumstances
- Typical: medical issues, mental health issues, interviews, emergencies
- We usually ask you to also inform your advisor

Group project

Instructor-assigned teams; assignments announced during Feb 2 lecture

Teams stay together for project throughout semester.

There will be a Catme Team survey around Jan 28 (3pt)

Some advice in lectures; we'll help with debugging team issues

TA assigned to each team as mentor; mandatory debriefing with mentor and peer grading on all milestones (based on citizenship on team)

Bonus points for social interaction in project teams

Academic honesty

- See web page; in a nutshell: do not copy from other students, do not lie, do not share or publicly release your solutions
- In group work, be honest about contributions of team members, do not cover for others
- *Do not falsely pretend to participate in in-class participation*
- Collaboration okay on labs, but not for individual assignments or exams
- If you feel overwhelmed or stressed, please come and talk to use (see syllabus for other support opportunities)

What makes software with ML challenging?

Lack of Specifications

```
/**  
    Return the text spoken within the audio file  
    ????  
*/  
String transcribe(File audioFile);
```

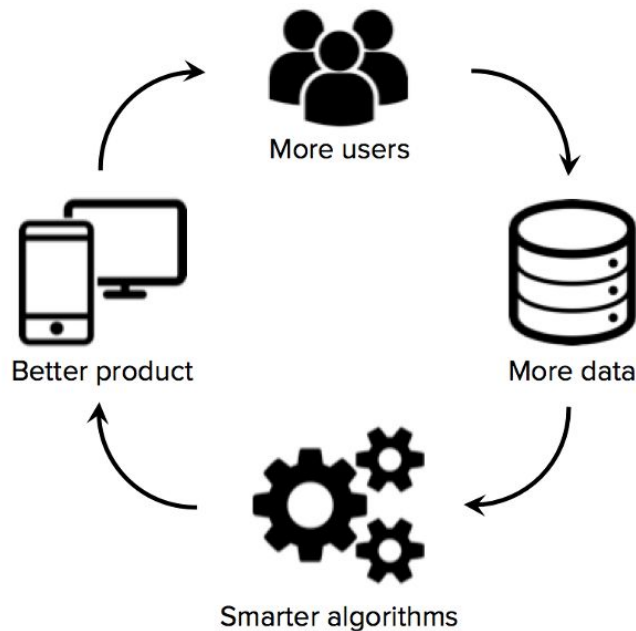
- Traditional SE: specify what to do & how to test
- ML usually black boxes (maybe justified, for their complexity...)
- Even if you put specs in, e.g., LLM prompts, unclear if they will follow
- Agents combine multiple unreliable components

Data Focused and Scalable

MLs get the “specs” from data; Larger the better

Deductive reasoning (applying logic rules) to
Inductive Reasoning (generalizing from observation)

Cause scalability issues



ML Models Make Mistakes

...often in unexpected ways

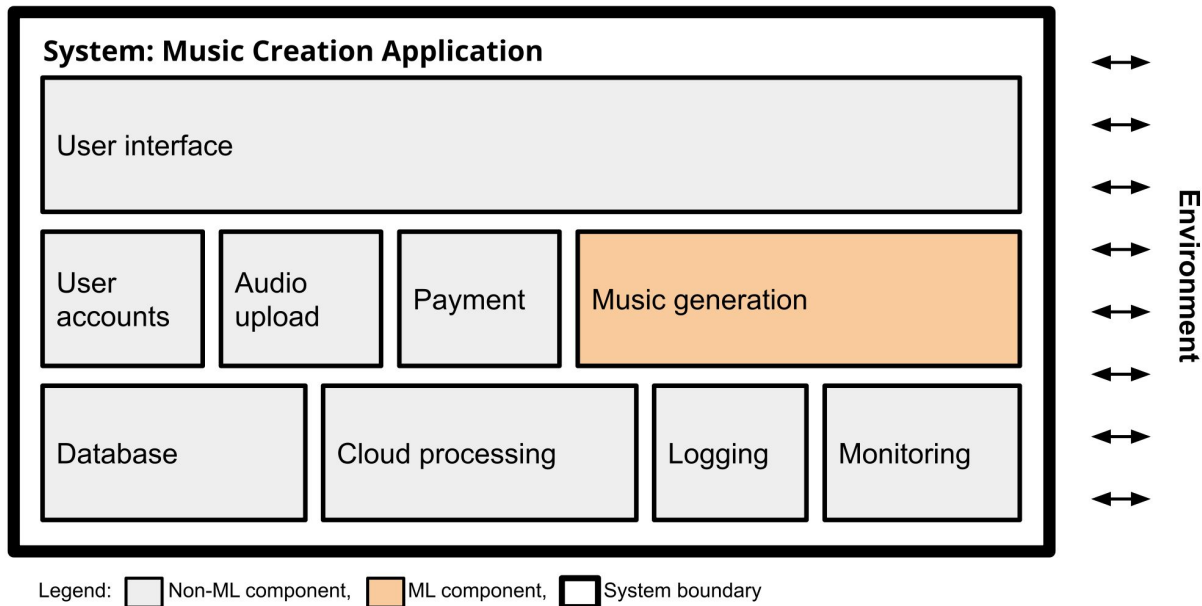
Hard to foresee and capture because no spec

What does it mean to be correct? Can only evaluate whether it works well enough (on average) on some test data!



NeuralTalk2: A flock of birds flying in the air
Microsoft Azure: A group of giraffe standing next to a tree
Image: Fred Dunn, <https://www.flickr.com/photos/gratapictures> - CC-BY-NC

Interaction with the environment



Our system must be able to tolerate some incorrect predictions, and be aware of how it might influence the world...

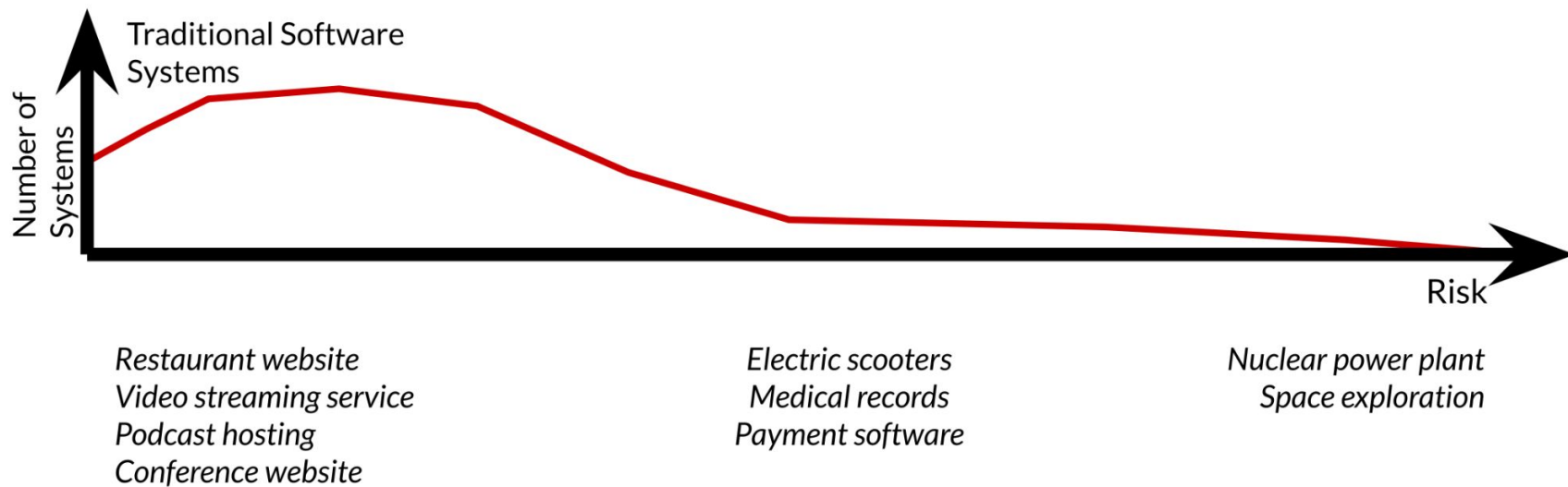
It's not all new

We routinely build:

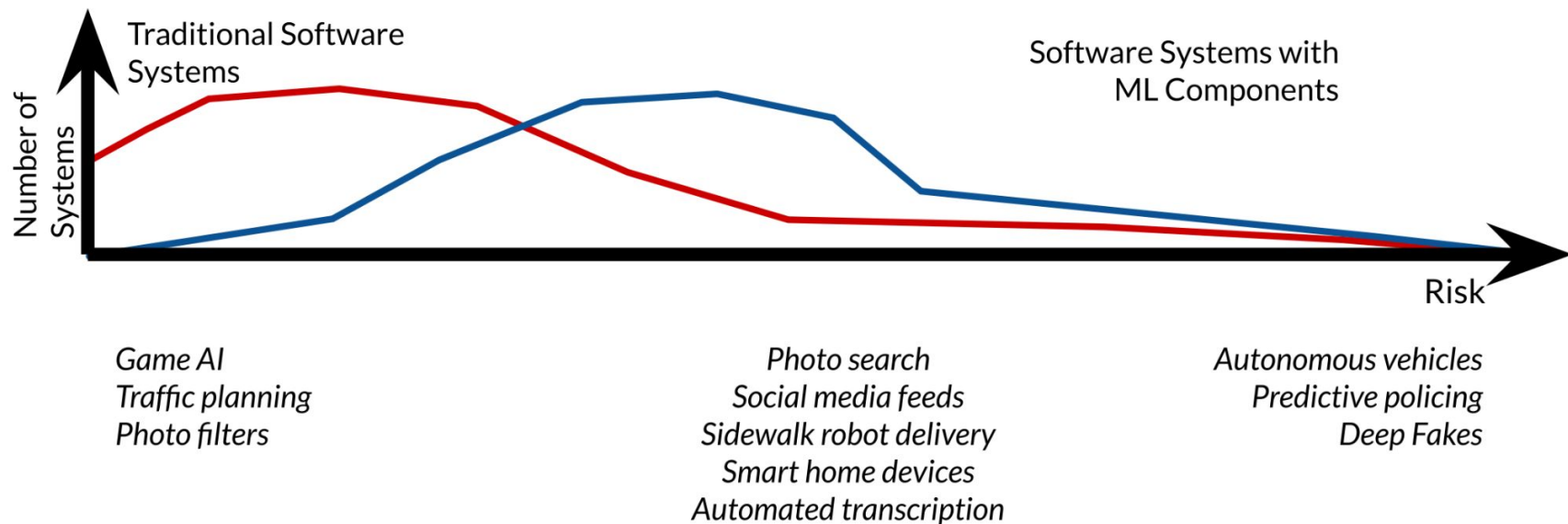
- Safe software with unreliable components
- Non-ML big data systems, cloud systems
- “Good enough” and “fit for purpose” not “correct”
- Cyberphysical systems

ML intensifies our challenges

Complexity



Complexity



Introductions

Before the next lecture, introduce yourself in Slack channel #social:

- Your (preferred) name
- In 1-2 sentences, describe your data science background and goals (e.g., coursework, internships, work experience)
- In 1-2 sentences, describe your software engineering background, if any, and goals (e.g., coursework, internships, work experience)
- The kind of job you are aiming for after graduation
- A hobby or a favorite activity outside school

Summary

Machine learning components are part of larger systems

Data scientists and software engineers have different goals and focuses

- Building systems requires both
- Various qualities are relevant, beyond just accuracy

Machine learning brings new challenges and intensifies old ones